

# Repository

## Repository Structure & Package Guide

This document describes the organization of directories, workspaces, and packages within the UnifyVault V2 monorepo workspace.

### Workspace Overview

UnifyVault is organized as a **pnpm monorepo** managed with **Turborepo**.

```
UnifyVault-UV/
├── apps/
│   ├── web-v2/           # Primary Next.js 15 production web application
│   └── web-v2-testnet/   # Staging/testnet environment deployment build
├── packages/
│   ├── protocol/        # Smart contracts, deployment scripts, & Foundry tests
│   ├── design-system/   # Shared React UI primitives & component library
│   ├── sdk/             # TypeScript client SDK for interacting with protocol contracts
│   ├── shared/          # Shared TypeScript interfaces, types, & constants
│   ├── eslint-config/   # Workspace-wide ESLint configuration package
│   ├── prettier-config/ # Workspace-wide Prettier formatting standards
│   └── tsconfig/        # Base TypeScript configuration templates
├── scripts/             # QA audit and operational automation scripts
├── docs/                # Complete protocol documentation suite
├── infra/               # Infrastructure configuration templates
├── pnpm-workspace.yaml  # pnpm workspace definition
└── turbo.json           # Turborepo task pipeline configuration
```

### Directory Descriptions

#### 1. `/apps`

Contains deployable applications:

- `apps/web-v2`**: The main user-facing decentralized web application (`app.unifyvault.xyz`). Built using Next.js 15 App Router, React 19, TailwindCSS, Wagmi v2, Viem v2, and RainbowKit v2. Features complete deposit/redeem flows, portfolio cost basis & P&L analytics, P2P Marketplace orderbook, and admin management dashboards.
- `apps/web-v2-testnet`**: Isolated staging environment build configured specifically for Base Sepolia testnet testing.

#### 2. `/packages`

Contains shared monorepo packages and smart contract suites:

- `packages/protocol`**: Core EVM smart contract implementation written in Solidity 0.8.24. Includes Foundry scripts (`script/`), unit/invariant/fork tests (`test/`), ABI artifacts (`out/`), and contract source code (`src/`).
- `packages/design-system`**: UI component library containing reusable atomic components (buttons, badges, modal dialogs, cards) standardizing visual presentation across apps.
- `packages/sdk`**: TypeScript SDK providing strongly-typed contract wrappers, NAV calculation helpers, and transaction builders for external integrations.
- `packages/shared`**: Shared type declarations, constants (e.g. BPS denominator, role hashes), and utility functions utilized by both frontend and SDK.

- `**`packages/eslint-config`**, **`packages/prettier-config`**, **`packages/tsconfig`**: Shared developer tooling configurations enforcing consistent code style across the monorepo.`

### 3. ``/scripts``

Contains automation scripts:

- `**`scripts/run_qa_audit.js`**`: Operational script performing static quality checks, ABI validations, and deployment state checks.

### 4. ``/docs``

Contains the complete, single-source-of-truth documentation set for the UnifyVault V2 protocol.

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## Workspace Scripts Summary

- **Command**: Workspace Target | Description
- ``pnpm build``: All Workspaces | Builds all smart contracts, packages, and Next.js applications via Turbo.
- ``pnpm dev``: ``apps/web-v2`` | Starts Next.js development server on port 3005.
- ``pnpm test``: ``packages/protocol`` | Runs Foundry smart contract test suite.
- ``pnpm format``: All Workspaces | Applies Prettier code formatting across all files.
- ``pnpm lint``: All Workspaces | Runs ESLint and Solhint across frontend and contract codebases.